CHEMISTRY: Sample Exam **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

SECTION I — MULTIPLE CHOICE

CALCULATORS CANNOT BE USED IN THIS SECTION

Each question is worth one point: one point for the correct answer. Completely fill in one bubble per question.

Questions 1 – 3 refer to the following terms.

(A) Electron Affinity
(B) Shielding Effect
(C) Ionization Energy
(D) Electronegativity
(E) Atomic Radius

1) A measure of the attraction of an atom for the electrons it shares in a chemical bond.

1 (A) (B) (C) (D) (E)

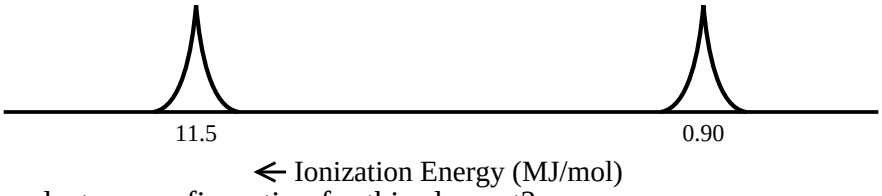
2) The energy change that occurs when a neutral atom in the gas phase gains an electron to form an anion.

2 (A) (B) (C) (D) (E)

3) One-half the distance between the nuclei of two identical atoms that are bonded together.

3 (A) (B) (C) (D) (E)

Questions 4 and 5 refer to the photoelectron spectrum below.



4) What is the electron configuration for this element?

(A) $1s^2 2s^2$ (B) $1s^2 2s^1$ (C) $1s^2 2s^2 2p^2$ (D) $1s^2 2p^2$ (E) $1s^1 2s^1$

4 (A) (B) (C) (D) (E)

5) Identify the element that would produce this spectrum.

(A) H (B) He (C) Li (D) Be (E) B

5 (A) (B) (C) (D) (E)

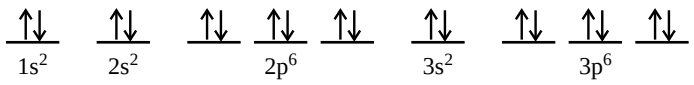
6) What mass of hydrogen gas is produced when 4.0 mol of sodium react with excess water according to the process outlined below?

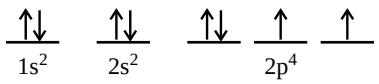
$$2\ Na + 2\ H_2O \rightarrow 2\ NaOH + H_2$$

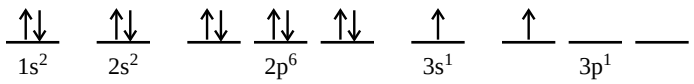
(A) 2.0 g H_2 (B) 4.0 g H_2 (C) 8.0 g H_2 (D) 16.0 g H_2 (E) 20.0 g H_2

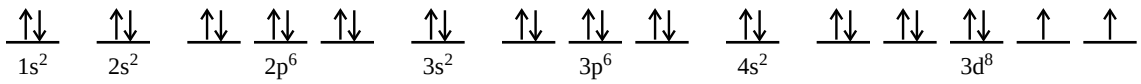
6 (A) (B) (C) (D) (E)

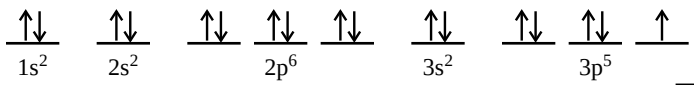
Questions 7 – 10 refer to the orbital diagrams below.

(A) 

(B) 

(C) 

(D) 

(E) 

7) Represents an atom with six valence electrons.

7 (A) (B) (C) (D) (E)

8) Does not violate any rules for filling orbitals and would not be attracted to a magnetic field.

8 (A) (B) (C) (D) (E)

9) Is in an excited state.

9 (A) (B) (C) (D) (E)

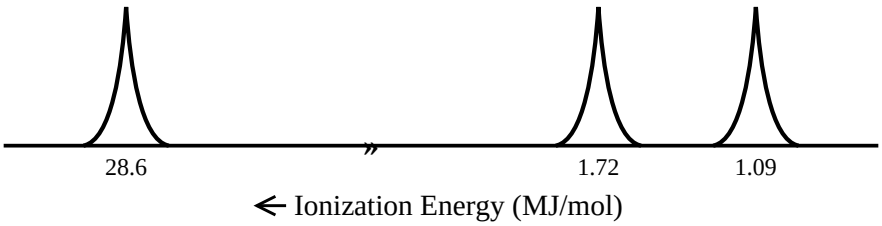
10) Represents a noble gas.

10 (A) (B) (C) (D) (E)

SECTION II — FREE RESPONSE

Show your work in the open area of each box; write only your final answer in the white FINAL ANSWER cell where provided. Point values for each part are printed beneath the question.

11) Use the photoelectron spectrum below to answer parts (a) – (d).



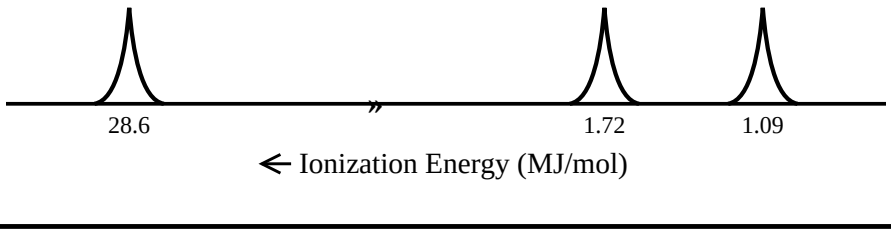
a. Identify the element that would produce the above spectrum.
One point for the correct answer.

11a

b. Label each peak on the spectrum with the numbers and letters that indicate its associated shell and subshell.
One point for the correct answer.

11b

LABEL THE SPECTRUM BELOW

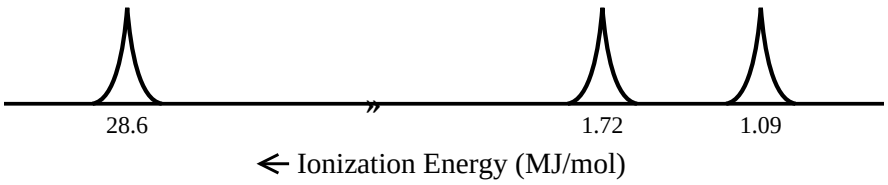


(Question 11, continued)

c. Indicate the number of electrons that are associated with each peak on the photoelectron spectrum.
One point for the correct answer.

11c

LABEL THE SPECTRUM BELOW



d. Identify the quantum level (e.g., n = 1, n = 2, …) that is holding the valence electrons in this element.
One point for the correct answer.

11d

12) Rb^{+} and Se^{2-} are isoelectronic.

a. Which ion has the smaller radius?
One point for the correct answer.

12a

(Question 12, continued)

b. Explain why the radii of these two ions are different sizes.
One point for the correct explanation.

12b

13) The following questions pertain to titanium.

a. Write the complete ground-state electron configuration for titanium, Ti.
One point for the correct answer.

13a

(Question 13, continued)

b. How many unpaired electrons does titanium have in its ground state? Justify your answer.

One point for the correct answer; one point for the justification. These points can be earned even if part (a) was incorrect, so long as your answer is consistent with the configuration you gave in part (a).

13b

JUSTIFY BELOW — ORBITAL DIAGRAM OR EXPLANATION

FINAL ANSWER NUMBER

c. Is a single gaseous titanium atom paramagnetic or diamagnetic? Justify your answer.

One point for the correct answer; one point for the justification.

13c

FINAL ANSWER PARAMAGNETIC / DIAMAGNETIC